

CHAPTER 3

Manufacturing Engineering

3.1. Introduction

Manufacturing engineering may be defined as *designing the production process for a product*. Although there is a large difference of opinion on exactly what is included in the design of a production process, almost all people would generally agree with the above definition. Production or manufacturing engineering includes all considerations pertaining to the *process* of production. This includes such functions as the following:

- (1) Evaluating the manufacturability of the product.
- (2) Selecting processes and setting process parameters such as cutting tool material, size, and shape; cutting speed; depth of cut; and so on.
- (3) Designing work-holding devices (jigs and fixtures) to secure and control the position of the workpiece during manufacture.
- (4) Estimating the cost of manufacturing the part.
- (5) Assuring the quality of the part produced.

The rate of technological change in manufacturing engineering is phenomenal and likely to increase. "Keeping up" is already a very difficult task. The future is likely to see a progressive manufacturing engineer spending 15 to 25% of his or her time simply studying new technology. Computerization, integration of controls,

robotization, and coordination of manufacturing activities are areas where change is likely to be most rapid. It is important to note, however, that in spite of the new technology, the basics still apply. For example, the most sophisticated computer-controlled machine center still performs its functions according to basic metal cutting theory. It is vital for the manufacturing engineer to know and understand the basics of manufacturing engineering.

This chapter discusses some of the primary areas of manufacturing engineering. Since much industrial engineering has been applied to the metal working industry, we concentrate on that particular industry. The reader should be aware that industrial engineering techniques can be applied in any operating system, such as manufacturing, service, and governmental activities. Fully 60% of the American work force functions in service and governmental areas; but the limited length of this chapter prohibits any detailed discussion of those areas.

3.2. Product-Production Design Interaction

Product design requires that a person develop and evaluate the ability of the part to perform its intended function. Part characteristics such as size, shape, strength, reliability, safe operating range, and so on, are evaluated using knowledge of physics, strength of materials, tribology, and so on, often using computerized analysis. Manufacturing engineering develops and evaluates the cost of producing the part and uses knowledge of the relative cost, capabilities, and limitations of the various alternative processing methods available to produce the particular part shape, as well as detailed knowledge of cutting tools, machine tools, skill levels of workers, similarity to other parts being produced, and so on. Unfortunately, few industrial engineers are trained in product design and few product designers are trained in manufacturing engineering. This makes their interaction exceptionally important.

In every manufacturing operation, some variation in the size of the individual parts produced by the process will inevitably occur due to a variety of causes, such as tool wear, operator error, and material variations. The range of part sizes that can be used without compromising part function or reliability, that is, the variation in each dimension of the product that can be tolerated, is referred to as the tolerance for the product. The part designer, who is most concerned with product function, would want the tolerances set as small as possible to assure that the part will function without any problems. The manufacturing engineer, who is most concerned with product cost, wants the largest possible tolerances to be specified, as this often gives him or her a wider choice of processes to be used in manufacturing the part. This wider choice will often result in a reduction in product cost. Sometimes, a product designer may specify very tight tolerances on the product because he or she does not realize the cost of machining to exceptionally tight tolerances or the inability of machines to produce unusual configurations. Often, the manufacturing engineer will assume that these tight tolerances or unusual configurations are necessary (when in fact they are not) and design the process to produce them. This adds unnecessary cost to the product.

Ideally, a manufacturing engineer should work with the product designer from the very beginning to ensure producibility. If it is not possible to have this early interaction, the manufacturing engineer should inform the design engineer of unusually costly operations. By having this information, the design engineer can frequently avoid certain costly operations. This interaction must happen and should occur as early as possible in the design of products.

Employee participation is also vital in this interaction phase. No one knows the details of a job better than the person doing it, so this employee feedback input must be encouraged for industry to remain competitive. In fact (as will be seen later), the use of employee skills in some type of participative management may be the most exciting trend today.

It is difficult to overemphasize the value of input from the hourly shop floor workers in planning and implementing automation systems. No one knows the nitty-gritty of a job better than the person who does it.¹

3.3. Process Engineering

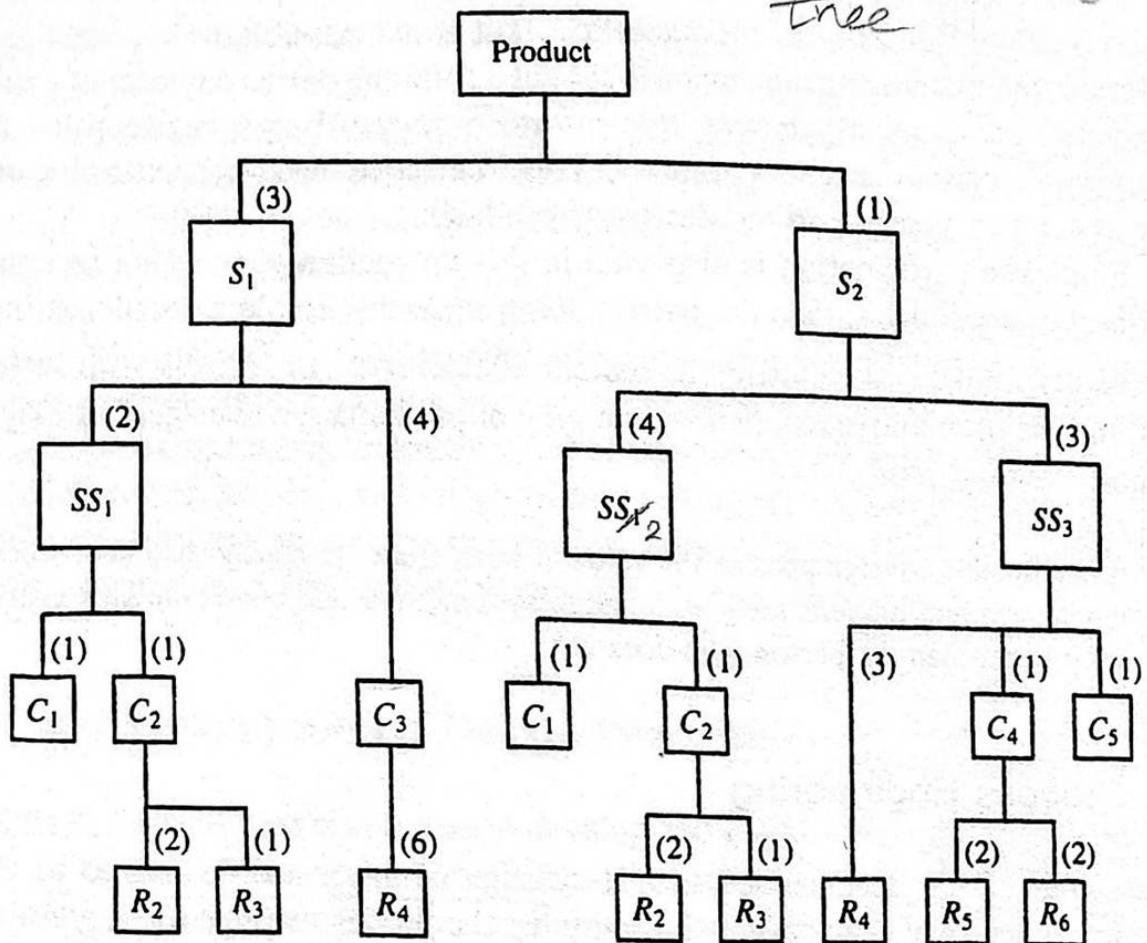
Process engineering is concerned with the design of the actual process to be used in the manufacture of the product. In designing the processes to be used, a six-step sequence should be undertaken: defining the product structure and specifications, assessing each component's manufacturability, listing the different processes capable to manufacturing the component, evaluating the cost of each of the alternative processes, determining the sequence in which the operations are to be performed, and documenting the process.

3.3.1 Defining Product Structure and Specifications

Product structures are often shown in a hierarchical chart that shows all of the subassemblies, sub-subassemblies, components, and raw materials that comprise the product. Figure 3.1 shows the product structure for a product assembled from two subassemblies (S_i), each of which is composed of sub-subassemblies (SS_j), components (C_k), and raw materials (R_l).

This type of chart clearly defines "what goes into what" and on which of the five levels the item belongs. More complex products would have more levels in their product structures. In Figure 3.1 each horizontal line indicates that an assembly operation must be designed to join its constituent parts. For example, the horizontal line linking subassemblies 1 and 2 indicates that a process to connect these two subassemblies must be designed. Note that sub-subassemblies, components, and/or raw materials can "go into" the subassembly; the hierarchy of the product structure dictates only that the horizontal line be above the highest-level item (subassembly,

¹Frank Curtin, "Automating Existing Facilities: GE Modernizes Dishwasher, Transportation Equipment Plants," *Industrial Engineering*, Sept. 1983, pp. 32-38.



* Fig. 3.1. Example product structure.

component, raw material) in the assembly. Also shown in Figure 3.1 are components that are purchased already fabricated and/or assembled and thus have no raw materials that must be processed by this facility. Vertical lines in the product structure indicate that the lower-level item is modified by some operation to transform or process it into the higher-level item. In industrial applications, each item in the product structure is identified by a unique number and incorporated in this number is the level of the item. This numbering is helpful when scheduling the production of the item, as we shall see in Chapter 7.

The number of each item required to make one of the next-higher-level item is also shown in the product structure. Figure 3.1 shows that three of subassembly 1 are required to make one unit of the product and that two sub-subassembly 1s are needed for each subassembly 1. This means that six units of sub-subassembly 1 are needed to make our product. These data are used when determining the number of each item that will be required to meet the sales forecast for our product.

This information can also be shown in bills of material. A bill of material contains the number and name of all parts, the source of each part (purchase or manufacture), and the number and name of the completed product. Normally, only those parts that go directly into that product are shown. For example, if a subassembly is used, the number and name of that subassembly are shown and a separate bill of material is developed for that subassembly. Figure 3.2 is a sample bill of material. In this figure, if the shelf (stock number 3) were a subassembly consisting

BILL OF MATERIAL			
Product Description: Bookcase, metal, 3 shelves. Stock Number: 1			
Component		Quantity (Amount) Required	Source
Stock No.	Description		
3	Shelf	3	Manufacturing
4	Leg	4	Manufacturing
5	Inserts	8	Purchasing
6	Screws	12	Purchasing

BILL OF MATERIAL			
Product Description: Bookcase, metal, 6 shelves. Stock Number: 2			
Component		Quantity (Amount) Required	Source
Stock No.	Description		
3	Shelf	6	Manufacturing
4	Leg	8	Manufacturing
5	Inserts	8	Purchasing
6	Screws	24	Purchasing
7	Connectors	4	Purchasing

BILL OF MATERIAL			
Product Description: Shelf Stock Number: 3			
Component		Quantity (Amount) Required	Source
Stock No.	Description		
8	Sheet metal	3 square feet	Purchasing

BILL OF MATERIAL			
Product Description: Leg Stock Number: 4			
Component		Quantity (Amount) Required	Source
Stock No.	Description		
8	Sheet metal	2 square feet	Purchasing

 **Fig. 3.2.** Bill of material. [Taken with permission, from Lawrence E. Doyle et al., *Manufacturing Processes and Materials for Engineers*, 2nd ed. (Englewood Cliffs, N.J.: Prentice Hall, 1969).]

of perhaps the basic shelf, a laminated walnut top, and molding, then the bill of material for the shelf would show these three items.

Nowhere have the powers of computers been more evident than in the generation of these forms (process sheets, route sheets, bills of material) and their subsequent use in production control. Many on-line computerized control systems generate the route sheets and trace the product as it flows through the shop. Recent research has even demonstrated the possibility of the computer going from product design through process engineering, through recognition of shapes and forms and "artificial intelligence." In artificial intelligence, the computer learns from past experiences until it actually is capable of designing the production process. The manufacturing engineer should watch this development carefully.

Once all of the items that "go into" the product have been enumerated using the product structure or bill of material, the specifications associated with each item must be developed. These specifications would detail the desired material, length, width, height, diameter, material, surface finish, and tolerances for the item, which will assure that it will perform its intended function. This will enable the evaluation of the item's manufacturability and development of a list of the processes that are capable of producing this item.

3.3.2 Assessing Manufacturability

→ standard-sized features
→ commonly available mat.

The manufacturability of an item is a measure of the relative ease or difficulty of producing the item within tolerance. The tighter the tolerances, in general, the more expensive the item will be to produce (see Figure 3.3). Also contributing to the manufacturability of the part is the incorporation of standard-sized features in the part. For example, if a hole in a part is specified to be a size that can be produced by a commonly available drill size, its cost to drill would be relatively low compared to the cost to drill a nonstandard hole size. Similarly, if at all possible, the part material should be specified to be a commonly available material. Only when particular part characteristics are not available from a commonly used material should an exotic material be specified. Each feature of each item in the product must be examined to determine if its function can be achieved in a manner that will allow consideration of a greater number of processing alternatives, some of which may be less expensive.

3.3.3 Determining Processes Capable of Producing the Part

By examining the specifications for each item and the capabilities of the various processes, a list of candidate processes can be developed. In developing this list, the following must be considered:

- (1) The compability of the material and the process.
- (2) The ability of the process to produce the desired tolerances.
- (3) Part (re)design required to make it easier to produce by this process.

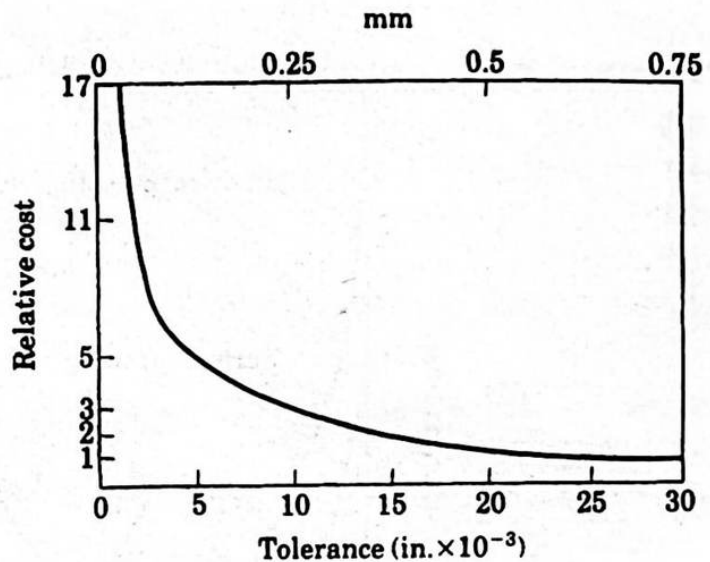


Fig. 3.3. Relationship of cost and tolerance. [From Serope Kalpakjian, *Manufacturing Engineering and Technology*, © 1989, by Addison-Wesley Publishing Company, Inc. Reprinted with permission of the publisher.]

- (4) The availability of the process.
- (5) The tooling, and so on, necessary to make the part using the process.

Use of tables such as those shown in Figures 3.4 through 3.6 enables the manufacturing engineer to quickly generate a complete list of available processes.

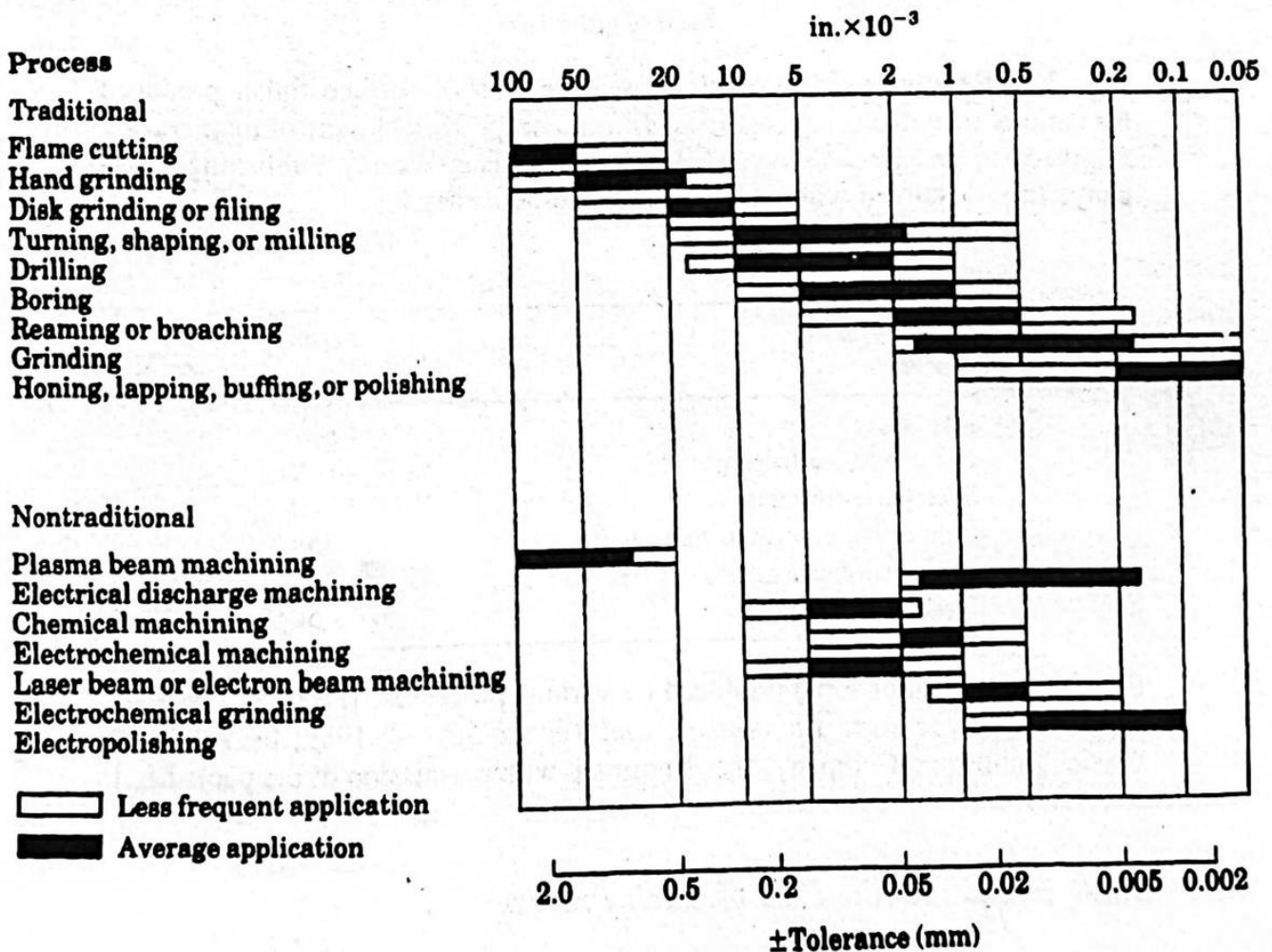


Fig. 3.4. Tolerances produced by various processes. [From Serope Kalpakjian, *Manufacturing Engineering and Technology*, © 1989, by Addison-Wesley Publishing Company, Inc. Reprinted with permission of the publisher.]

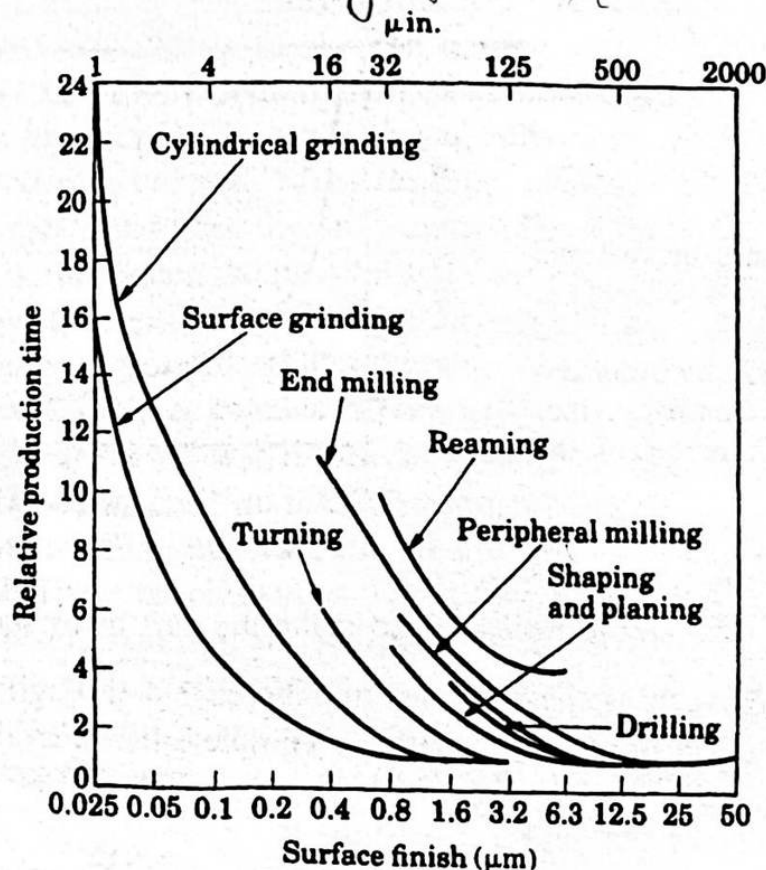


Fig. 3.5. Relative production time as a function of surface finish produced by various manufacturing methods. [From Serope Kalpakjian, *Manufacturing Engineering and Technology*, © 1989, by Addison-Wesley Publishing Company, Inc. Reprinted with permission of the publisher.]

Process	Scrap (%)
Machining	10–60
Hot closed-die forging	20–25
Sheet-metal forming	10–25
Cold or hot extrusion, forging	15
Permanent-mold casting	10
Powder metallurgy	5

Fig. 3.6. Amount of scrap produced by various processes. [From Serope Kalpakjian, *Manufacturing Engineering and Technology*, © 1989, by Addison-Wesley Publishing Company, Inc. Reprinted with permission of the publisher.]

3.3.4 Evaluating the Cost of Each Process

To evaluate the cost of using each process, two types of costs must be determined: fixed costs and variable costs. Fixed costs are those costs that are constant regardless of the number of parts made using the process. Examples of fixed costs are the purchase cost of the machine(s) used in the process, the cost of

installing the machine(s), the cost of designing and fabricating the work-holding devices, and the cost of the space that the machine(s) occupy. Variable costs are those costs that vary directly (i.e., proportionally) with the number of parts produced. Examples of variable costs include the cost of the machine operator's time while making the part, the cost of running the machine(s) (e.g., electricity, natural gas, other fuels), the cost of all cutting tools that are dulled or otherwise consumed during the manufacture of the part, and the cost of the material used in manufacturing the part.

Fixed costs can be estimated by contacting vendors and/or users of the various equipment that is being considered as well as cost data developed by the firm's accounting department. Estimates of the variable cost rate (i.e., dollars/hour) can be obtained from the same sources. Expressing this in an equation, we obtain

$$\text{total cost} = \text{fixed cost} + \text{variable cost/unit} \times \text{number of units}$$

This relationship is also shown in Fig. 3.7.

If a number of alternative processes are being considered, their costs can be plotted on the same graph and the range of production quantities for which each process is less expensive can be determined. For example, suppose that three processes are under consideration and the following costs apply:

	Process A	Process B	Process C
Fixed cost	\$ 0	\$2,000	\$8,000
Cost/unit	10	8	5

If we graph the costs, as shown in Figure 3.8, we can see that if we need to produce fewer than 1,000 units, process A should be used. If more than 2,000 units are needed, process C should be used. For production requirements between 1,000 and 2,000, process B should be used. The production requirement of 1,000 is said to be the break-even point between process A and process B and is the production

Comparison?
depends on
the # produced

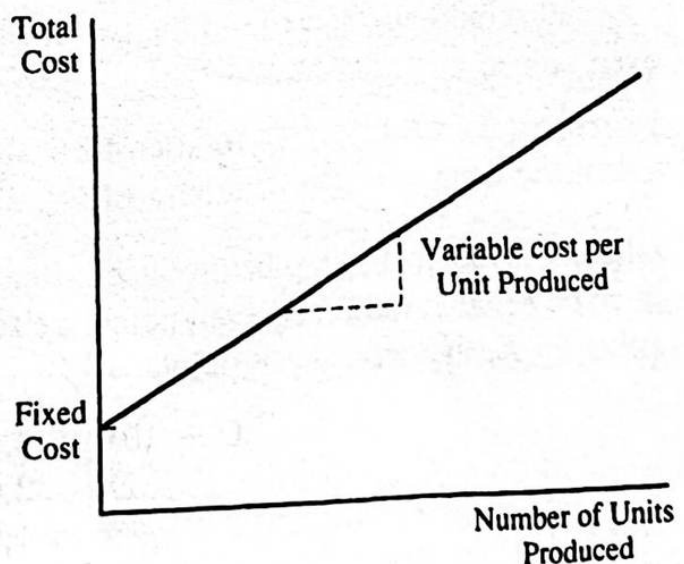


Fig. 3.7. Total cost as a function of the number produced.

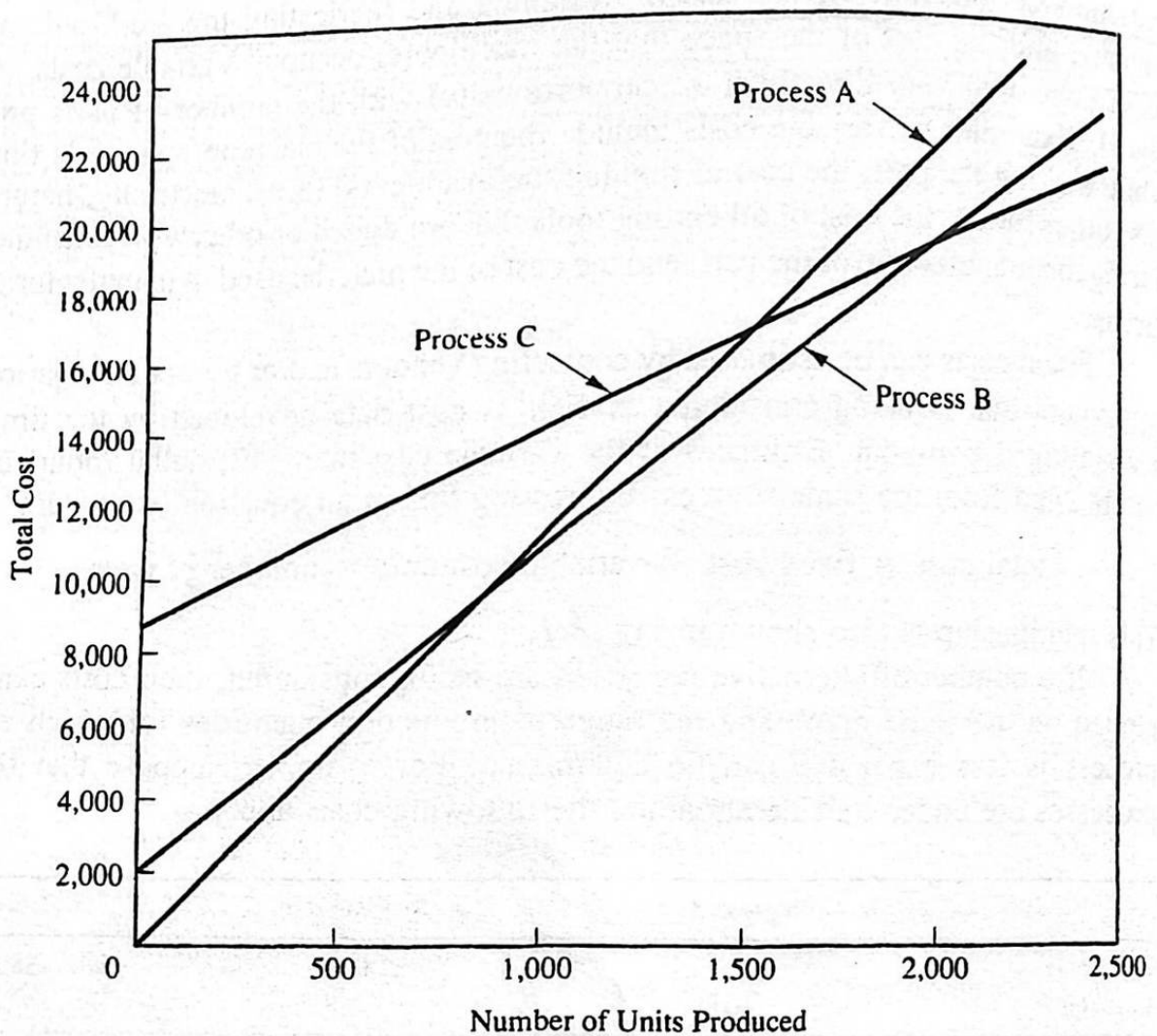


Fig. 3.8. Total cost versus quantity for three processes.

volume at which the total manufacturing costs are equal for the two processes. Production quantities below this break-even point can be produced at a lower cost using process A, and quantities above this break-even point can be produced at a lower total cost using process B. Similarly, 2,000 is the break-even point for processes B and C.

We could also have used the two total cost equations to determine the break-even points:

$$\text{total cost}_A = 0 + 10X$$

$$\text{total cost}_B = 2,000 + 8X$$

where X is the quantity to be produced. Realizing that total costs for both processes must be equal at the break-even point, we set the equations equal to each other and solve for X , the break-even point.

$$0 + 10X = 2,000 + 8X$$

$$2X = 2,000$$

$$X = 1,000$$

Similar calculations can be used to verify that the break-even point between process B and process C is 2,000.

The variable cost per part reflects the time that it takes to manufacture one part using the process and the costs that are incurred during this period of time.² This variable cost for a metal cutting operation has four components: the cost of labor (operator) and the machine while the machine is cutting, the cost of the tool, the cost of labor and the machine while sharp tools are being mounted, and the cost of labor and the machine while the workpiece is being loaded and unloaded. These costs can be expressed as

$$\begin{aligned} \text{variable cost} = & T_c \times r_{l,m} + r_t \times \frac{T_c}{T} + T_{dt} \times r_{l,m} \\ & \times \frac{T_c}{T} + r_{l,m} \times T_{dw} + r_{l,m} \times T_a \end{aligned}$$

where T_c = cutting time/part (min)

T = tool life (i.e., amount of time a tool can cut before dulling) (min)

$r_{l,m}$ = cost (dollars/min) of running the machine, including operator cost

T_{dt} = time (min) it takes to put a sharp tool on the machine (downtime for tool change)

r_t = cost of one sharp tool

T_{dw} = time (min) to load/unload one workpiece (downtime for workpiece change)

T_a = time (min) to adjust or position tool on each workpiece

This cost is dependent on the process parameters that are selected for the operation. The need to define the relationship of the cutting speed and tool life was addressed by Frederick W. Taylor, the father of industrial engineering, early in the twentieth century.³

He knew that the time it takes to machine a part can be determined by calculating the length of the cut(s) that are required to machine the part to the desired size. For example, if a shallow groove is to be cut into a 36-inch-long workpiece using a single-point cutting tool operating at 20 ft/min, it would take 9 seconds to complete the cut. If 20 grooves were required on the part, 3 minutes of cutting would be required to manufacture the part. If one wanted to reduce the processing time, one could increase the cutting speed. The problem with this approach is that when the cutting speed is increased, the tool wears out or dulls much more quickly. To determine the optimum cutting speed, the relationship between the speed at which a cutting tool was being operated and how long the tool would cut until it was too dull to use was needed. This relationship is described by the following formula, known as *Taylor's tool life equation*:

²Vidosic, Joseph P., *Metal Machining and Forming Technology* (New York: Ronald Press, 1964), p. 329.

³Taylor, F. W., "On the Art of Cutting Metals," *Transactions of ASME*, 28 (1906), p. 31.

$$C = VT^n$$

where C = cutting speed (ft/min) at which the tool would dull after one minute of use

V = cutting speed (ft/min) at which the process will be operated

T = tool life (min) that can be expected when the tool is operated at cutting speed V

n = a constant that is dependent on the material being cut and the cutting tool material

If we define the length of the cuts on a particular workpiece to be L , $T_C = L/V$, and if we rearrange the terms in the Taylor tool life equation,

$$T = \left(\frac{C}{V}\right)^{1/n}$$

we can substitute these two variables into our variable cost equation:

$$\begin{aligned} \text{cost/part} = & \frac{L}{V} \times r_{l,m} + r_t \times \frac{L}{V} \times \left(\frac{C}{V}\right)^{-1/n} \\ & + T_{dt} \times r_{l,m} \times \frac{L}{V} \times \left(\frac{C}{V}\right)^{-1/n} + r_{l,m} \times T_{dt} + r_{l,m} \times T_a \end{aligned}$$

Since this is now an equation involving only the single variable V , we can differentiate it with respect to V , set the first derivative equal to 0, and solve for V . When we do,

$$V = C \left(\frac{n}{1-n} \right)^n \left(\frac{r_{l,m}}{r_{l,m}T_{dt} + r_t} \right)^n$$

which is the optimum cutting speed, that is, the cutting speed that minimizes the cost of cutting each workpiece and hence minimizes the cost of producing the product on the machine in question.

Suppose, for example, that a 2-foot section of a 3-in.-diameter 1020 steel part is to be machined using a high-speed steel tool. Assume that:

$$\begin{array}{lll} C = 225 \text{ ft/min} & r_{l,m} = 1 \text{ \$/min} & r_t = \$4 \\ n = 0.105 & T_{dt} = 3 \text{ min} & \end{array}$$

$$\begin{aligned} V &= 225 \left(\frac{0.105}{1-0.105} \right)^{0.105} \left[\frac{1}{(1)(3) + 4} \right]^{0.105} \\ &= (225)(0.7976)(0.8152) \\ &= 146.2 \text{ ft/min} \end{aligned}$$

Further, the time to cut the part using a feed rate (the translation of the tool per revolution of the workpiece) of 0.05 in./rev:

$$\text{No. revolutions to cut 24 in.} = \frac{24}{0.05} = 480$$

$$\text{rev/min at cutting speed of 146 ft/min} = \frac{146 \text{ ft/min}}{\pi \left(\frac{3}{12} \right) \text{ ft/rev}} = 186 \text{ rev/min}$$

$$\text{time to cut } (T_c) = \frac{480}{186} = 2.58 \text{ min}$$

$$\text{tool life } (T) = \sqrt[n]{\frac{C}{V}} = 0.105 \sqrt{\frac{225}{146}} = 61.5 \text{ min}$$

$$\text{length of the cut} = \pi(3) \left(\frac{24}{0.05} \right) = 4,523 \text{ in.} = 377 \text{ ft.}$$

and if $T_{dt} = 3 \text{ min}$, $T_{dw} = 1 \text{ min}$, $r_{l,m} = \$1/\text{min}$, $T_a = 2 \text{ min}$, and $r_t = \$1/\text{edge}$:

$$\begin{aligned} \boxed{\text{total cost}} &= \frac{L}{V} r_{l,m} + r_t \frac{L}{V} \left(\frac{C}{V} \right)^{-1/n} \\ &\quad + T_{dt} r_{l,m} \frac{L}{V} \left(\frac{C}{V} \right)^{-1/n} \\ &\quad + r_{l,m} T_{dt} + r_{l,m} T_a \\ &= \$7.87 \end{aligned}$$

3.3.5 Determining the Sequence of Operations

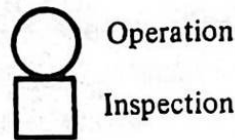
The determination of the sequence in which the operations that will be used to transform the material into its desired final shape depends on a number of considerations, and a full discussion of them is beyond the scope of this book. The student should be aware, however, that the sequence is developed based on (1) minimization of part handling (the part is routed, if possible, along the shortest path from machine to machine without backtracking); (2) assuring that no succeeding operation adversely affects previous operations by leaving burrs, debris, scratches, and so on; and (3) performing as many operations on each machine as possible so that close tolerances can be maintained and quality assured.

3.3.6 Documenting the Process

After the manufacturing engineer has reviewed the various processes and has decided on the most economical and effective procedure or set of processes, he or she must then find a way of recording this information for later use in the shop. One way is to construct an operations process chart.

An operations process chart depicts the flow of the material through the various processes. It shows only operations and inspections; it does not show transportations, storages, or delays. These are shown in flow process charts that will be discussed

later. Using the following symbols, a manufacturing engineer can construct an operations process chart (see Fig 3.29 for an example.)



In the operations process chart, a complete unit is shown on a page (whenever possible). The chief component starts at the top and subsidiary components are added as the flow progresses to the right and down. (This will be discussed again in Chapter 5 and an example will be presented later in this chapter.)

The next step is to develop a route sheet that shows the sequence of operations, a description of each, and the various machining information necessary. A route sheet is a set of instructions to the plant showing how to manufacture the product. Figure 3.9 is an example of a route sheet. One such sheet would be required for each part in the operations process chart. The exact information shown varies between companies, but a good rule of thumb is to provide enough information to be able to produce the part including machine numbers, tool numbers, directions, and perhaps times for setup of the equipment and actual run time required. Figure 3.9 is not only a routing sheet, but also a production order for use in dispatching an order to the shop.

* 3.4. Industrial Processes

There is a wide variety of processes available to the manufacturing engineer to transform material into the desired size and shape. While a large number of materials are used in products, we will limit our discussion to those used for production of metal parts. These processes include refining and alloying of the metal, casting, forming, metal cutting, welding, assembly, and finishing.

3.4.1 Refining and Alloying

The general procedure for almost all metals is to begin with ore in its natural state and send it through some process to improve its usefulness. Steel, for example, is extracted as iron ore, is sent through a blast furnace to produce pig iron, and then it is sent to the steel mill where it is made into steel by the open hearth, basic oxygen, or electric furnace process. The type of steel produced is determined by the temperature of the furnace and its chemical composition.

Since all metals do not have the same properties, the proper choice of the metal to be used in a product depends on the functions the part must perform. This choice occurs in the production-product design interaction. To further compound the problem, metals emerging from the primary operations usually do not have the properties necessary. This means that some sort of metallurgical transformation is

TR		W-WORK ORDER R-ROUGH STORES F-FINISHED STORES		PRODUCTION WORK ORDER		MIN. LOT 72 PART NO. 8B-3381	
TOTAL COPIES		SHEET NO.		EFFECTIVE DATE IF NOT STARTED			
MACHINE		BASE TIME		OPERATION			
OPER. NO.	LOC. CLASS	M. NO.	HRS. PER SET-UP	HRS. PER PIECE	BURDEN HRS.	PERFORM. HRS.	
1	X18B A2C	1939	1.00	.0009 1111			CHAMFER & CUT OFF
2	X23D G2R	1383	.10	.0025 400			REMOVE BURRS
3	X19C G20C	1997	.50	.0061 164			GRIND O.D. (2) PASSES
4	X19C M1B	7080	.70	.0161 62			STRADDLE MILL TONGUE
5	X19C M1B	7080	.80	.0141 71	.80 .0141	.80 .0141	MILL KEYWAY
6	X23D G2R	1383	.10	.013 77			REMOVE BURRS & CHAMFER TONGUE
7	X19C D6S	6076	.30	.0167 60			DRILL HOLE
8	X22D E1B	B131	.20	.0050 200			REMOVE BURRS
PARTS GROUP NO. 49							
FILE	HT	I	C	W	R	F	
COMP. PARTS		MATERIAL					(N.C.)
4B-588		C.T. #1E-40 C.F. STEEL 5/8" RD. M.L.					
4B-608		PART NAME					QTY. PER PC. .505 FT.
4B-3429		OIL PUMP GEAR SHAFT					
7B-4920							
3B-6794							
4B-8735							
GG:MKG:CK		ORDER DATE					DELIVER TO
		8B-3381					SCHED. LOT SIZE
							SCHEDULE
							CSTGB. REQ'D
		PART NO.					

Fig. 3.9. Sample route sheet. [Taken with permission, from Lawrence E. Doyle et al., *Manufacturing Processes and Materials for Engineers*, 2nd ed. (Englewood Cliffs, N.J.: Prentice Hall, 1969).]

required to improve the hardness, strength, workability, or other properties of the metal. Much of this is done by heat treating the metal to alter its properties. In some cases, it may be necessary to combine two or more metals to obtain better properties. These combinations are called *alloys*. The production engineer must be aware of the different types of metals available and the particular alloys that can be made and their properties.

Because the properties of the various metals are very important, testing for these properties is of primary importance. Some of the properties that are often tested are:

- (1) *Tensile Strength*—The ability of the metal to withstand elongation forces.
- (2) *Hardness*—The ability of the metal to withstand penetration forces.
- (3) *Impact Resistance*—The ability of the metal to absorb energy.
- (4) *Malleability*—The ability of the metal to be shaped or extended.
- (5) *Fatigue Resistance*—The ability of the metal to withstand repeated applications of a force.
- (6) *Corrosion Resistance*—The ability of the metal to withstand corrosive forces.

The production engineer must understand these and other tests in order to select the correct material.

3.4.2 Casting

The first step in production is to obtain an approximate shape that, if necessary, can later be refined or finished more smoothly. One way of doing this is through casting. Casting is the process of forming objects by pouring liquid material into a mold and allowing the material to solidify. Casting is an old process, but it is still the most economical and effective method for obtaining many desired shapes.

Mold design is one of the most important parts of casting and the correct design of the mold is both a science and an art. Although many types of molds exist, by far the most predominant is a sand mold, especially for iron and steel.

Figure 3.10 illustrates a typical sand mold with its principal parts. Each part is described below.

Flask—Frame in which the mold is made.

Cope—Top section of the flask.

Drag—Bottom section of the flask.

Cheek—Intermediate sections of the flask (when required).

Riser—Reservoir of liquid metal to supply additional liquid metal as the metal solidifies and contracts.

Pouring Basin—Area into which molten metal is poured.

Sprue—Channel for vertical delivery of molten metal.

Runner—Channel for horizontal delivery of molten metal.

Gate—Opening through which metal enters the mold.

The correct relative and absolute shapes and dimensions of each of the above parts are highly critical in the design of a good mold. If the manufacturing engineer

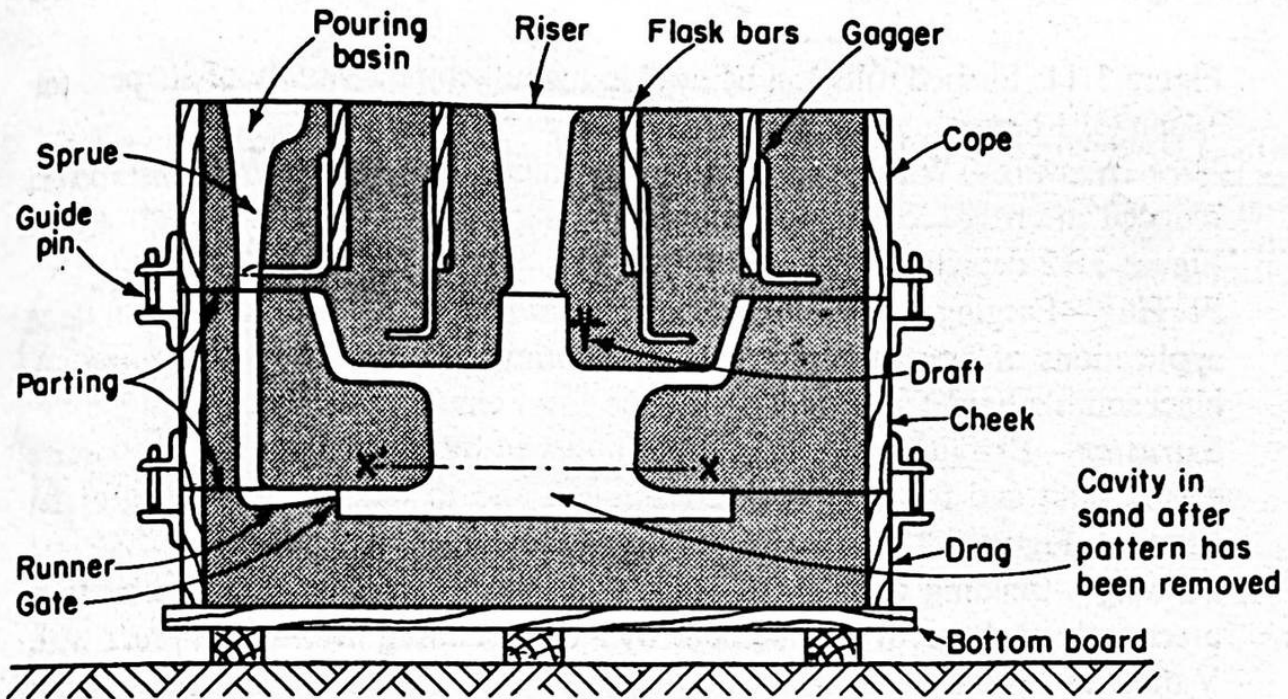


Fig. 3.10. Cross-sectional view of a three-part mold. [Taken with permission, from Lawrence E. Doyle et al., *Manufacturing Processes and Materials for Engineers*, 2nd ed. (Englewood Cliffs, N.J.: Prentice Hall, 1969).]

is to be working with castings, he or she should be cognizant of these and should be able to design good molds.

Obviously, this type of sand casting is expensive when large quantities of products are to be made, as the mold is destroyed each time a casting is made. Consequently, many specialized types of casting with permanent molds have been developed. The subject is too long and complex for this book, but some of the specialized types are centrifugal casting, die casting, gravity or permanent mold casting, pressed casting, ceramic shell casting, plaster mold casting, and shell molding casting.

3.4.3 Metal Forming

Another type of primary metal working process is metal forming. Here, the metal is worked under pressure to form a desired shape, to improve the physical properties, or both. This working can be either *hot* or *cold*. In hot working, the metal is worked at a temperature above its recrystallization temperature so that certain properties are enhanced. For example, in hot working, the metal working requires less pressure, and unusual shapes can be made more easily. Cold working takes place at a temperature below the metal's recrystallization temperature. This is done to hold close tolerances, to produce good surface finishes, and to enhance other physical properties.

Some of the processes that are classified as metal forming operations are listed below with short descriptions. Many of these processes can be performed in either hot or cold states, depending on the needs.

Rolling—Rolling is a squeezing operation in which the metal is elongated and/or widened as it passes through two or more rollers, as depicted in

Figure 3.11. Shaped rolls can be used to manufacture a variety of shapes, for example, I-beams.

Wire Drawing—Wire drawing is an operation in which a wire or rod is reduced in cross-sectional diameter by being drawn (pulled) through a die. Figure 3.12 depicts the process.

Forging—Forging is metal forming by a single or a series of intermittent applications of pressure instead of by continuous pressure as in rolling. A blacksmith's hammering of a horseshoe is an example of forging.

Extrusion—Extruding a metal is accomplished by compressing it beyond its elastic limit and forcing it to flow through and to take on the shape of an opening. Figure 3.13 shows examples of extrusions.

Bending—Bending is applying force to permanently distort the metal to a preconceived shape. It may be done by a tube bending machine, a press with V dies, or several other means. Figure 3.14 depicts some typical sheet metal bends.

Drawing and Stretching—These operations produce seamless vessels by applying pressure and forcing the sheet metal to take on the shape of a preconceived form. As in wire drawing, some stretching of the material occurs. Figure 3.15 is an example of a simple drawing operation.

3.4.4 Metal Cutting

After the primary shape forming, the next step is often metal cutting. Metal cutting is the removal of metal by using a cutting tool. The objective is to obtain desired shapes, close tolerances, and specified surface finishes. There are many types of metal cutting operations. Some of the better known are listed below with short descriptions:

Shearing—Shearing is the cutting of sheet metal by applying pressure and forcing the metal between two sharp edges. Blanking, parting, punching,

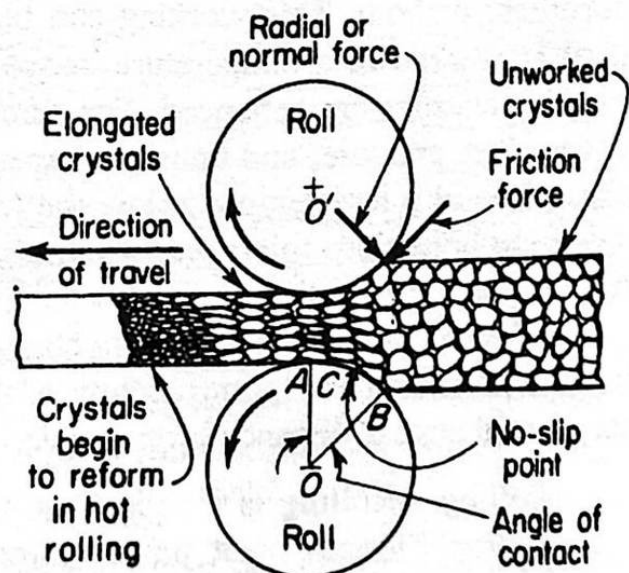


Fig. 3.11. Rolling operations. [Taken with permission, from Lawrence E. Doyle et al., *Manufacturing Processes and Materials for Engineers*, 2nd ed. (Englewood Cliffs, N.J.: Prentice Hall, 1969).]

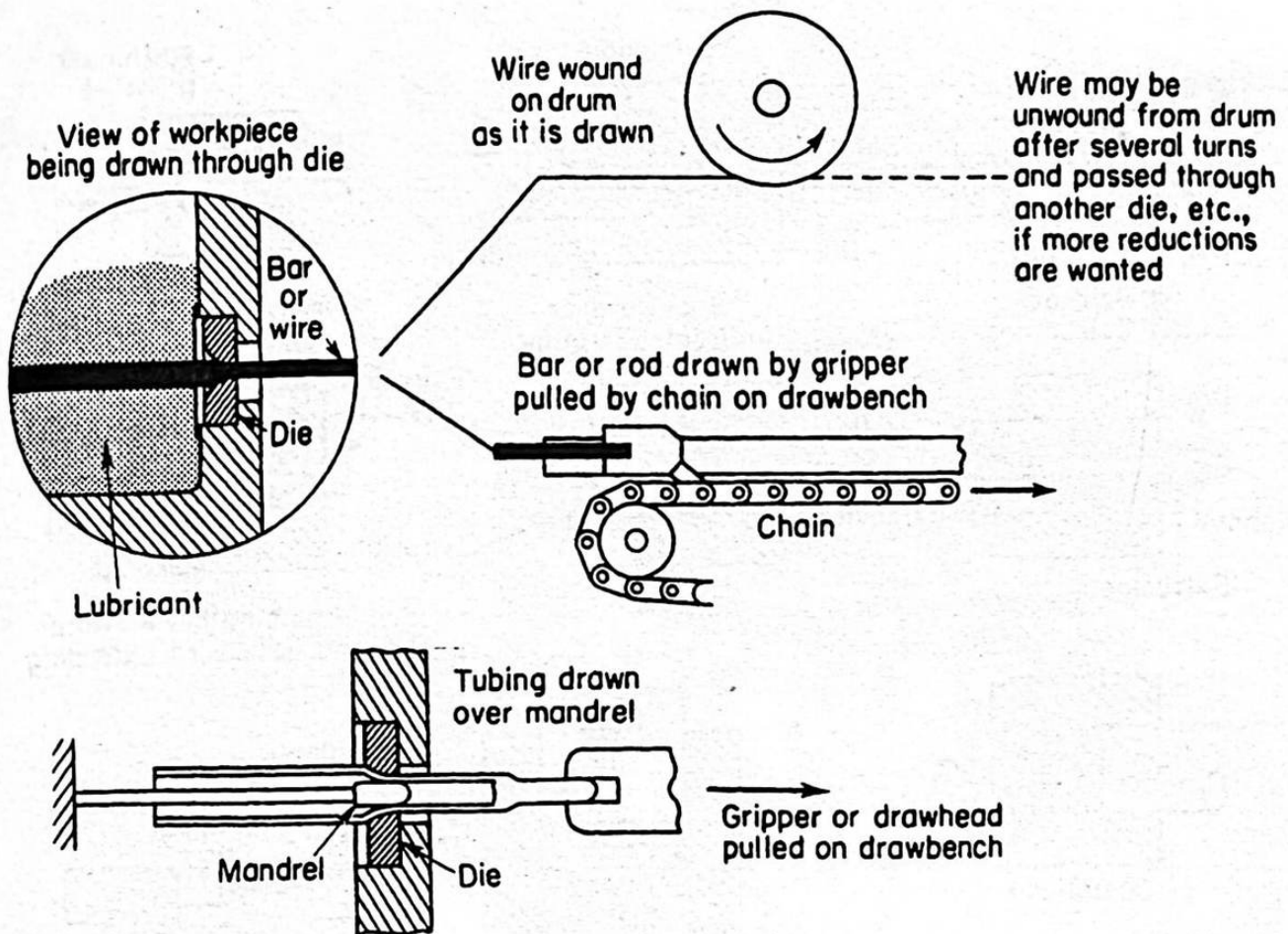


Fig. 3.12. Drawing operations. [Taken with permission, from Lawrence E. Doyle et al., *Manufacturing Processes and Materials for Engineers*, 2nd ed. (Englewood Cliffs, N.J.: Prentice Hall, 1969).]

nibbling, notching, and shearing along a straight line are all shearing operations. Examples are shown in Figure 3.16.

Turning—Turning is the process of rotating the workpiece and cutting material by feeding a cutting tool into the workpiece and/or the workpiece into the tool. These operations are generally performed on a lathe and may include facing, contour turning, threading, parting, drilling, knurling, and others.

Drilling—Drilling is the opening, enlarging, or finishing of a hole. It is generally associated with the process in which the tool turns and the workpiece is stationary. When the workpiece turns, it is known as a *turning operation* (see preceding page). The operation is generally performed on a drill press or boring machine.

Shaping and Planing—These are operations in which surfaces are cut by a reciprocating action. Generally, the surfaces are flat, but curved surfaces can also be machined by the processes. In shaping, the tool reciprocates while the workpiece is stationary. In planing, the workpiece reciprocates while the tool is stationary. Figure 3.17 is an illustration of a shaper in operation.

Milling—Milling is the process by which a revolving cutting tool with a number of teeth takes intermittent, successive cuts. Usually the workpiece is fed into the tool, but this is sometimes reversed. Figure 3.18 depicts some common milling operations.

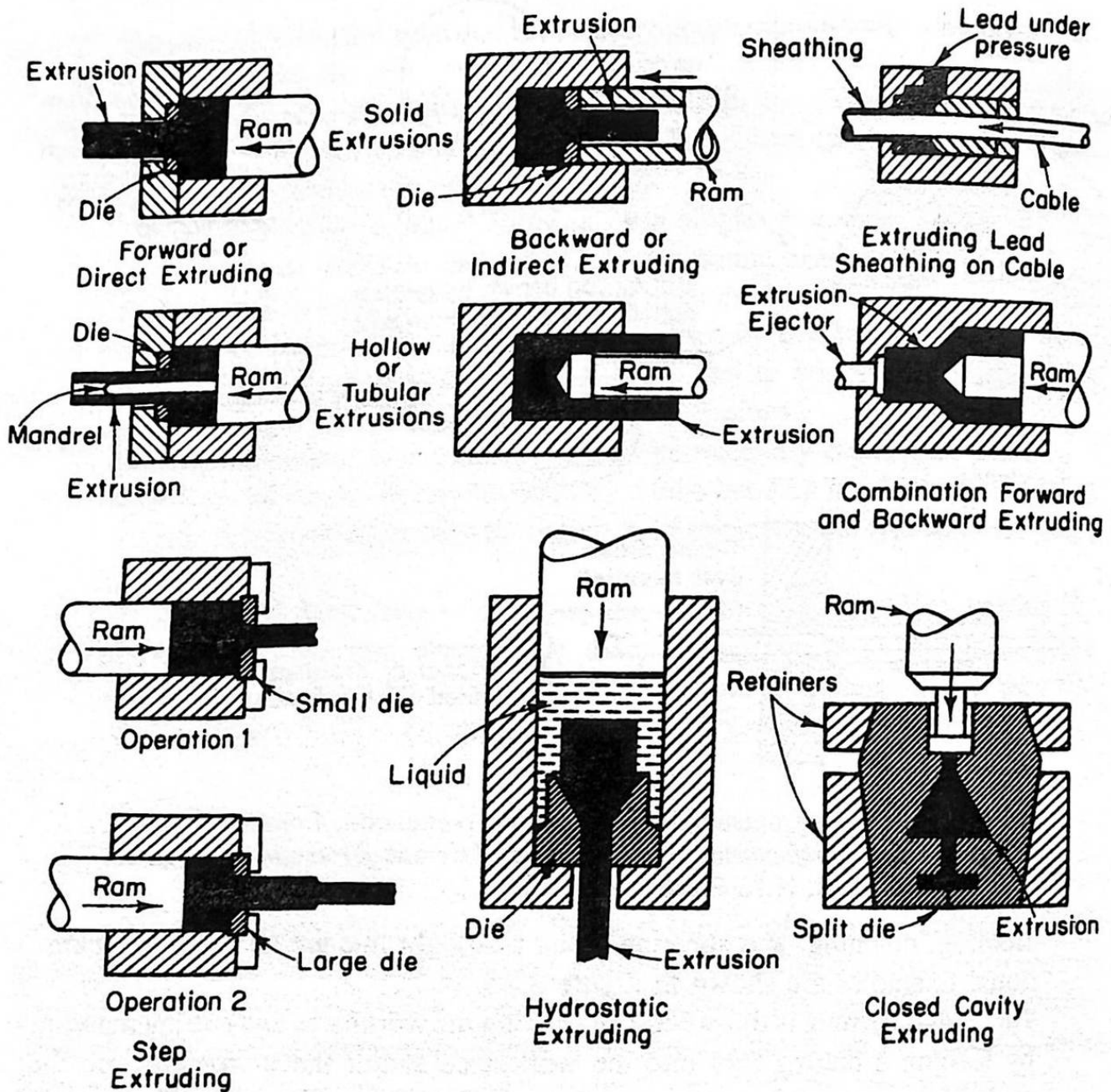


Fig. 3.13. Extruding. [Taken with permission, from Lawrence E. Doyle et al., *Manufacturing Processes and Materials for Engineers*, 2nd ed. (Englewood Cliffs, N.J.: Prentice Hall, 1969).]

Broaching—Broaching is similar to milling in that a series of teeth are used to take successive cuts on the workpiece. The broaching tool does not revolve, however; it is pushed or pulled. Figure 3.19 shows the basics of a broaching operation.

Sawing—Filing—Sawing and filing also remove metal by taking cuts with successive teeth on a tool that is pushed or pulled. In sawing, however, large amounts of metal can be removed without reducing it to chips. Figure 3.20 is a picture of a power hacksaw in operation.

Grinding—Grinding is the removal of metal in small pieces by the continuous action of an abrasive material on the workpiece. Grinding may be necessary on materials too hard for other methods, to improve surface finishes, or to hold very close tolerances. The small grinder in a garage used to sharpen hand tools is a familiar example.

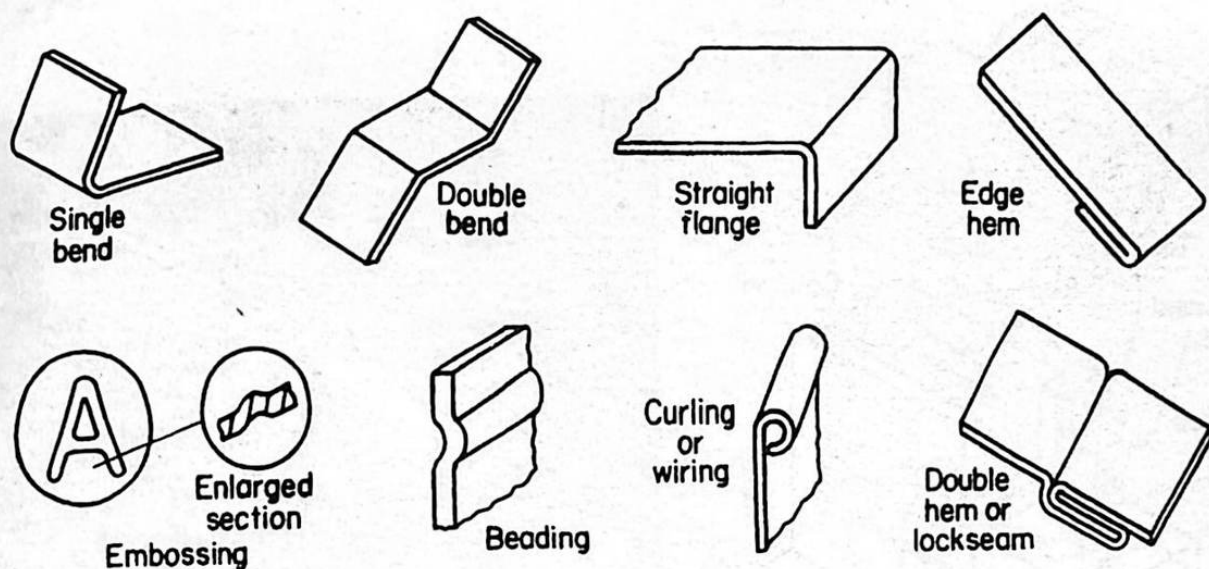


Fig. 3.14. Examples of sheet metal bends. [Taken with permission, from Lawrence E. Doyle et al., *Manufacturing Processes and Materials for Engineers*, 2nd ed. (Englewood Cliffs, N.J.: Prentice Hall, 1969).]

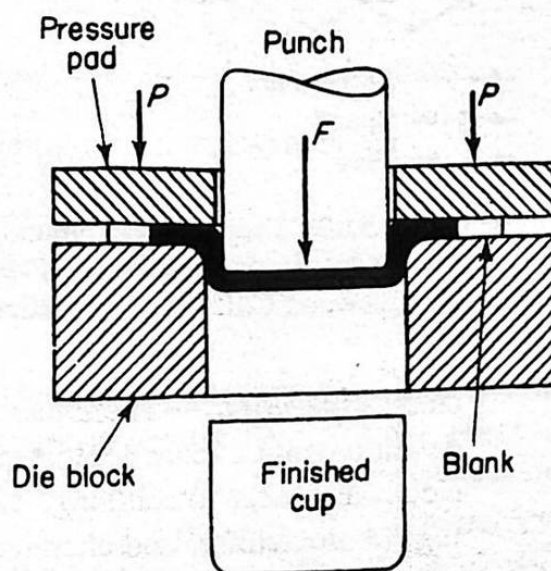


Fig. 3.15. Drawing of a cup. [Taken with permission, from Lawrence E. Doyle et al., *Manufacturing Processes and Materials for Engineers*, 2nd ed. (Englewood Cliffs, N.J.: Prentice Hall, 1969).]

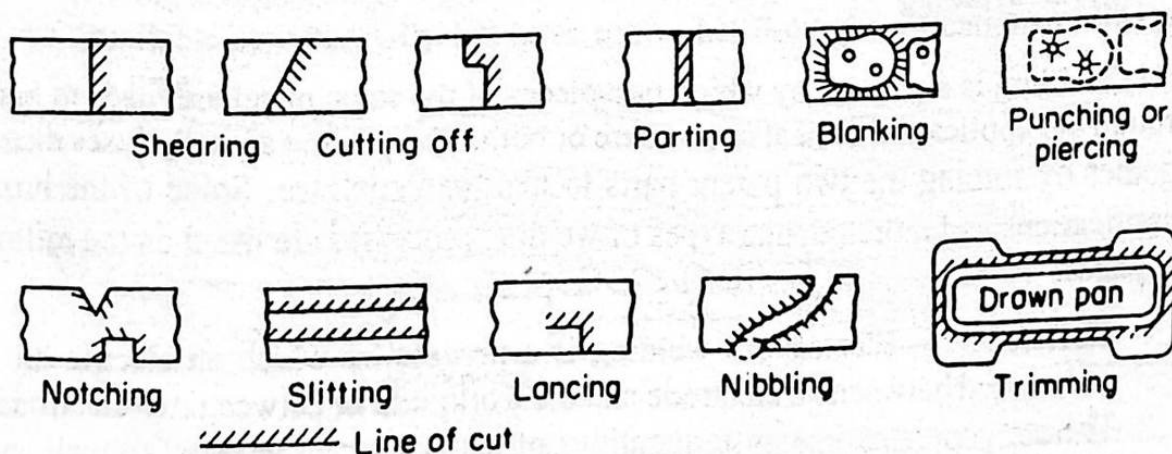


Fig. 3.16. Typical shearing operations. [Taken with permission, from Lawrence E. Doyle et al., *Manufacturing Processes and Materials for Engineers*, 2nd ed. (Englewood Cliffs, N.J.: Prentice Hall, 1969).]

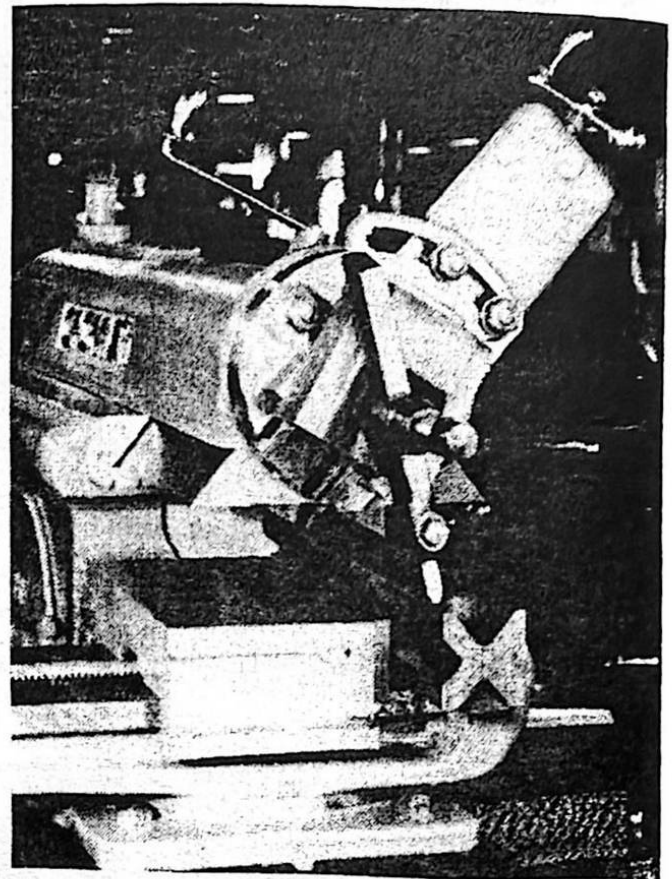
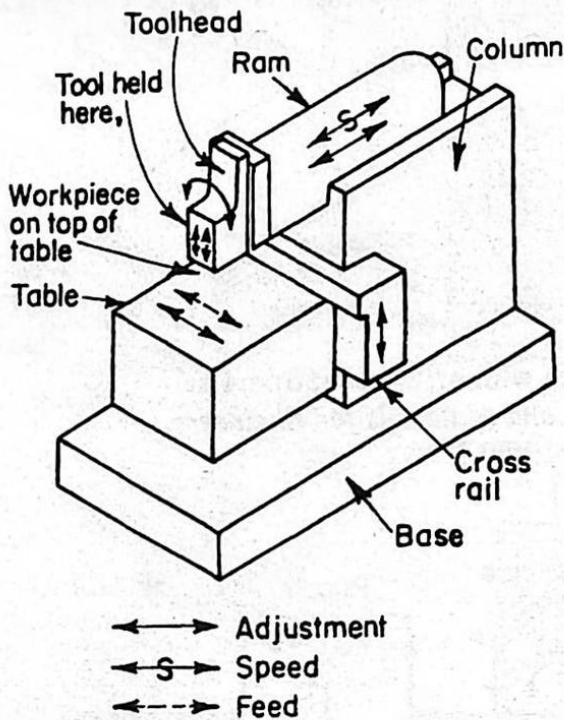


Fig. 3.17. Shaper in operation. [Taken with permission, from Lawrence E. Doyle et al., *Manufacturing Processes and Materials for Engineers*, 2nd ed. (Englewood Cliffs, N.J.: Prentice Hall, 1969).]

Other Operations—There has been a great deal of work directed toward the development of more sophisticated methods of metal removal. Examples are electrodischarge machining, electrochemical machining, electrolytic grinding, photoetching, and chemical milling. These will not be discussed in this chapter.

3.4.5 Welding

Welding is a process by which two pieces of the same metal are made to bond through the application of heat or pressure or both. The process actually fuses metals together by forcing the two parent parts to melt and coalesce. Some of the broad classifications and more popular types of welding processes are listed on the following pages.

Electric Arc—Electric arc welding is a process by which an electric arc is maintained between an electrode and the workpiece or between two electrodes. The arc generates immense quantities of heat causing the metal to melt and, upon cooling, cause the weld to occur. In some cases, the electrode provides the filler metal and sometimes a slag or protective covering over the weld.

Resistance Welding—In resistance welding, an electric current is passed

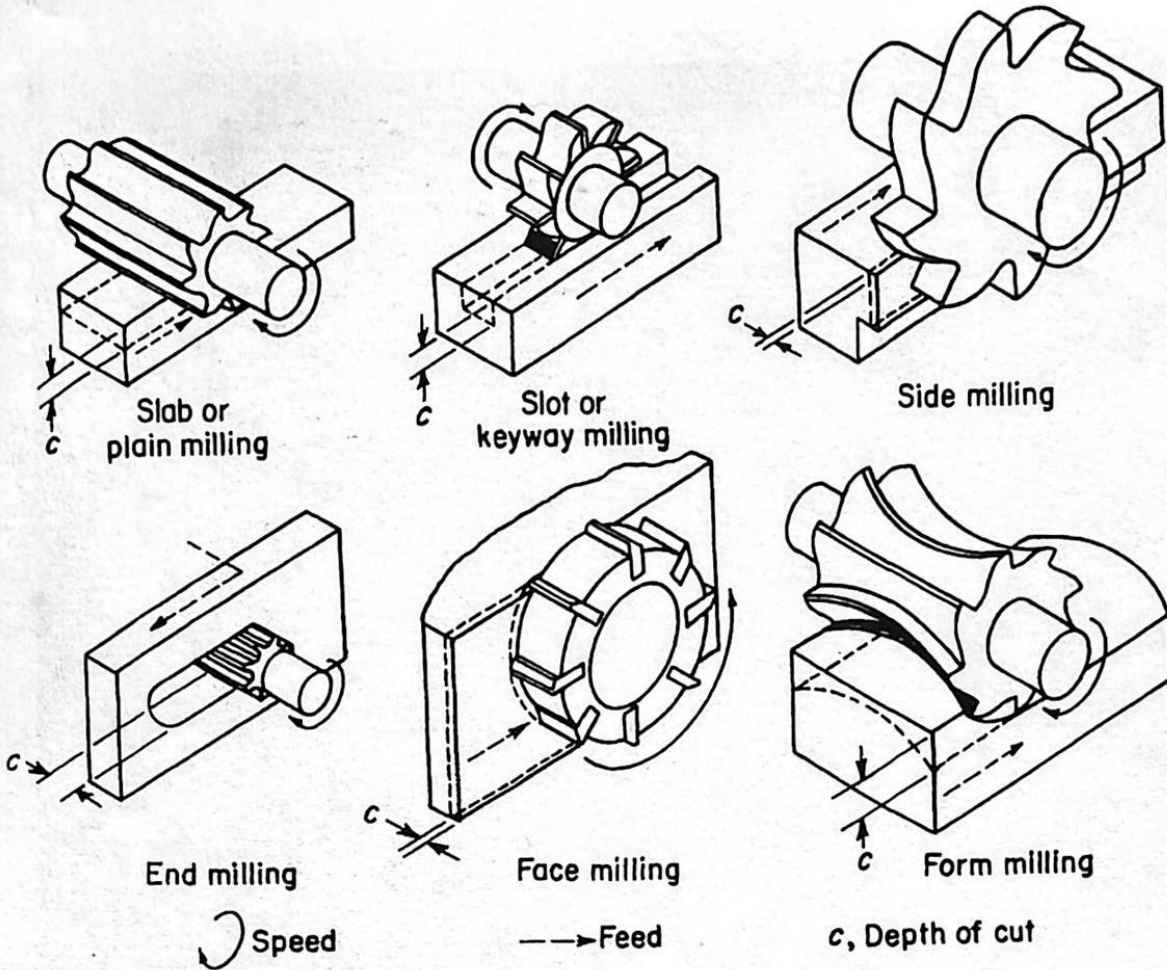


Fig. 3.18. Common milling operations. [Taken with permission, from Lawrence E. Doyle et al., *Manufacturing Processes and Materials for Engineers*, 2nd ed. (Englewood Cliffs, N.J.: Prentice Hall, 1969).]

through two pieces of metal that are held together under pressure. The electric resistance across the surface discontinuity generates heat and the pressure causes the two metals to bond. Spot and seam welding are two popular examples of resistance welding. Figure 3.21 is a diagram of a spot welder.

Beam Welding—This process supplies the heat by bombarding the workpiece with a concentrated beam of electrons (electron beam welding) or by concentrating a high-energy light beam on the workpiece (laser welding). Both are available commercially but have somewhat limited applications at this point in time.

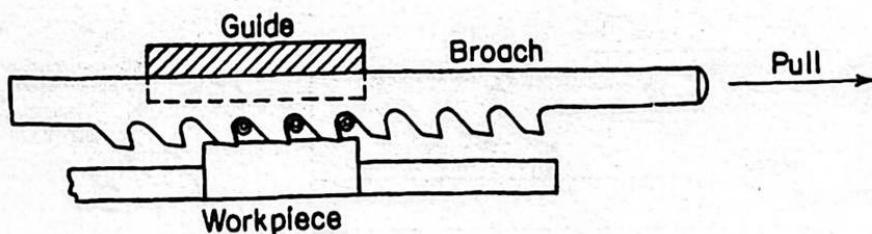


Fig. 3.19. Broaching operation. [Taken with permission, from Lawrence E. Doyle et al., *Manufacturing Processes and Materials for Engineers*, 2nd ed. (Englewood Cliffs, N.J.: Prentice Hall, 1969).]

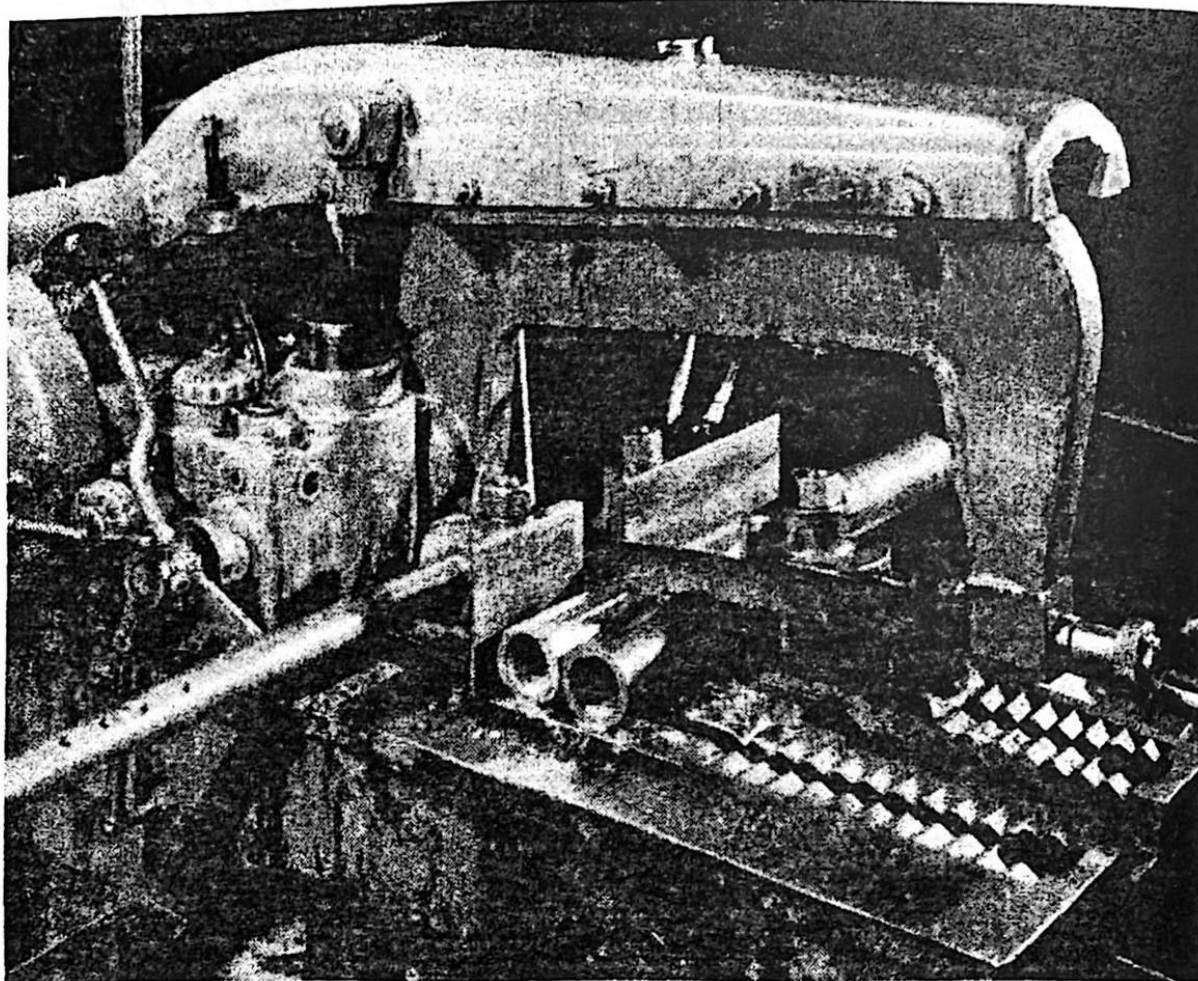


Fig. 3.20. Power hacksaw. [Taken with permission, from Lawrence E. Doyle et al., *Manufacturing Processes and Materials for Engineers*, 2nd ed. (Englewood Cliffs, N.J.: Prentice Hall, 1969).]

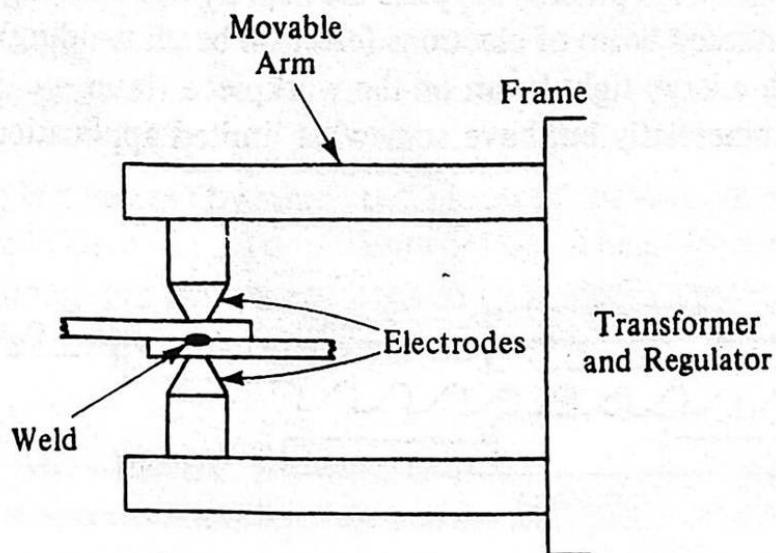


Fig. 3.21. Diagram of a spot welder. [Taken with permission, from Lawrence E. Doyle et al., *Manufacturing Processes and Materials for Engineers*, 2nd ed. (Englewood Cliffs, N.J.: Prentice Hall, 1969).]

Thermit Welding—Thermit welding is a process by which a cavity is formed by the pieces to be welded and molten metal poured into the cavity to make the weld. The molten metal is usually iron, steel, or copper and the welding process is particularly useful for large parts.

Pressure Welding—In pressure welding, the two parts are made to join by applying sufficient pressure with or without an external heat source. The pressure must be high enough to *squeeze* out all imperfections. Friction welding is a good example of pressure welding.

Gas Welding—In gas welding, a combustible gas is burned with air or oxygen in a concentrated flame. This flame is used to heat the parent pieces and filler for the actual weld.

Brazing and Soldering—In brazing and soldering, the filler metal is melted and added to the weld area. The filler is chosen so that its melting point is below that of the parent metal. Consequently, the parent pieces never melt, and when the filler material cools and solidifies, it acts as a (metallic) glue to hold the parts together.

3.4.6 Assembly

Assembly is the process by which the various parts and subassemblies are brought together to form a completed assembly or product. Since there are numerous methods of assembly operations, they will not be covered in detail.

For example, welding is sometimes called an assembly process although assembly usually implies less permanent bonds. Gluing, riveting, screwing, and force fitting are familiar examples of assembly processes. The manufacturing engineer must be familiar with these various processes and understand the relative advantages and disadvantages of each process.

Organizationally, the assembly may be done in a *batch* process whereby assemblies are made intermittently or in a *continuous* process such as assembly lines. The manufacturing engineer must be familiar with these concepts; he or she must know how to set up and balance assembly lines, and so on.

3.4.7 Finishing

Finishing is a set of processes in which the material, either as a completed product, subassembly, or component part, is made more effective or presentable through the external application of energy and/or other materials.

Honing and lapping are two processes in which an abrasive material is rubbed over the metal to produce a fine surface. Honing and lapping are similar to grinding except that the speed is slower and the part is never overheated. The purpose is to improve the surface finish. Other more familiar finishing operations are polishing, buffing, brushing, and tumbling.

Another form of finishing is surface cleaning and coating. The purpose of cleaning is to remove dirt and scale that may be contaminating the workpiece,

usually prior to coating. Familiar cleaning operations are spraying, vapor degreasing, and dip baths (such as pickling with an acid solution).

The purpose of coating is to provide decoration, protection, texture, and other desirable properties. Painting and plating are two of the most common coating techniques. Metallizing (spraying of the surface with an atomized molten metal) and phosphatizing and chromating (formulation of a noncorrosive surface over the metal that is also a good paint bond) are examples of less familiar coating operations.

The manufacturing engineer must be familiar with these cleaning and coating techniques and the advantages and disadvantages of each. He or she must know how to set up the processes and estimate the cost of each.

3.5. Ancillary Functions

The production or manufacturing engineer performs many other functions other than the process selection just discussed. Some of these are tool, fixture, and jig design, cost estimating, maintenance systems design, and packaging systems design. Each of these is discussed below.

3.5.1 Tool, Jig, and Fixture Design

Tool design is the designing of tools for the most effective and economical operation. Although it is one of the most important areas of production-manufacturing engineering, tool design is overlooked (except in passing) by almost all industrial engineering curricula because it is so large and complex (and often an art as opposed to a science) that it would be very difficult to cover this subject adequately. Correct die design for a stamping operation could be a course in itself. Obviously, it would be practically impossible to thoroughly cover tool design for all metalworking operations, not to mention the basic operations in other industries.

Figure 3.22 gives an indication of the many parameters involved in designing a very simple cutting tool. Each of the angles shown must be selected based on the material being cut, the material of the cutting tool, and the cutting conditions. Changes of one degree in these angles can significantly affect the cutting operation. Imagine, then, the complexities and problems in designing a progressive (several stages or operations sequentially performed by one die) blanking, piercing, and forming die such as shown in Figure 3.23. Many industrial engineers become tool designers who learn almost all their skills on the job.

Similarly, jig and fixture design is an important phase of production-manufacturing engineering passed by in many industrial engineering curricula for the same reasons discussed above. A fixture is a device used to hold and locate a workpiece. The vise found in home workshops is a simple example of a fixture. In highly repetitive and exacting work, fixture design can be very important. A jig is similar to a fixture except that it also guides the tool into the workpiece. Figure 3.24 is an

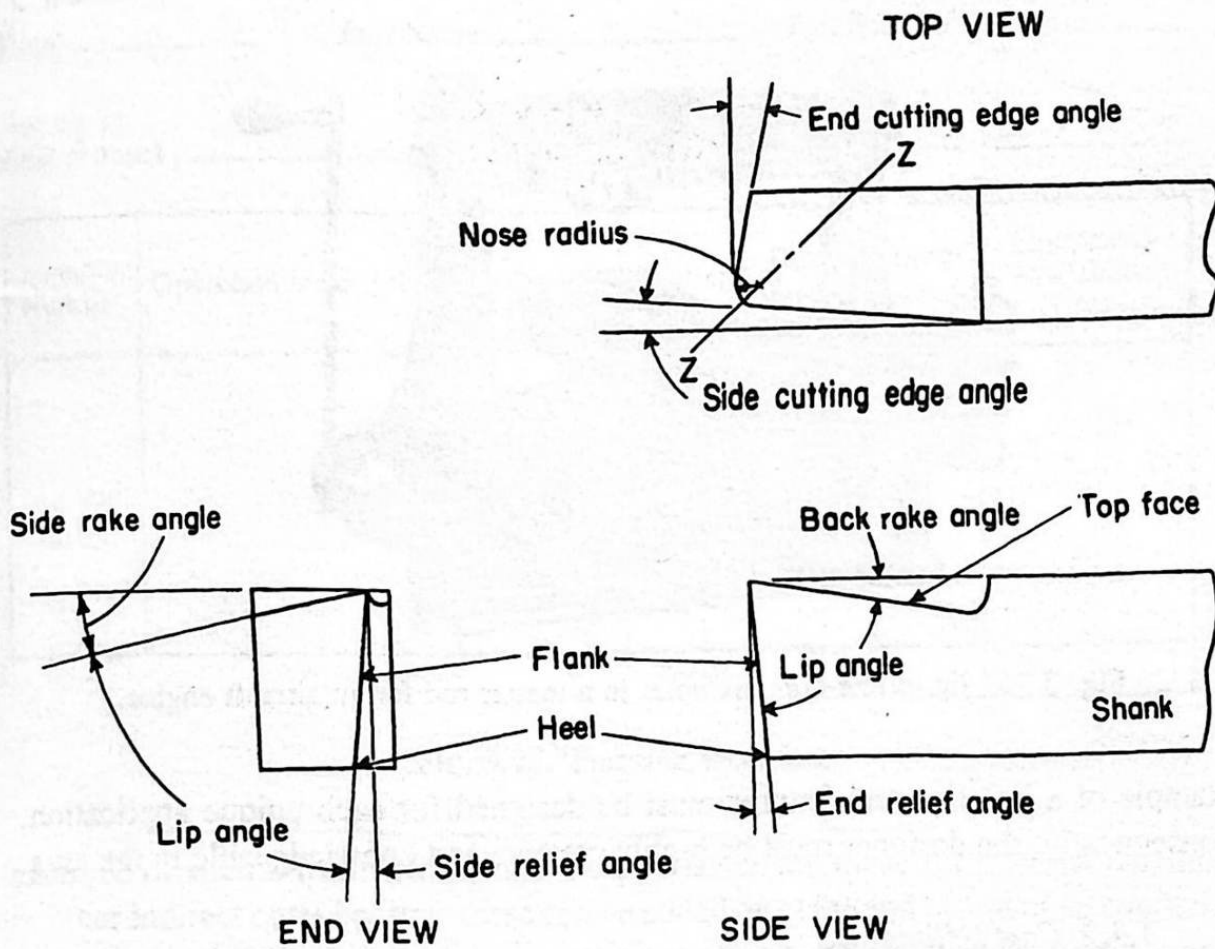


Fig. 3.22. Cutting tool design considerations.

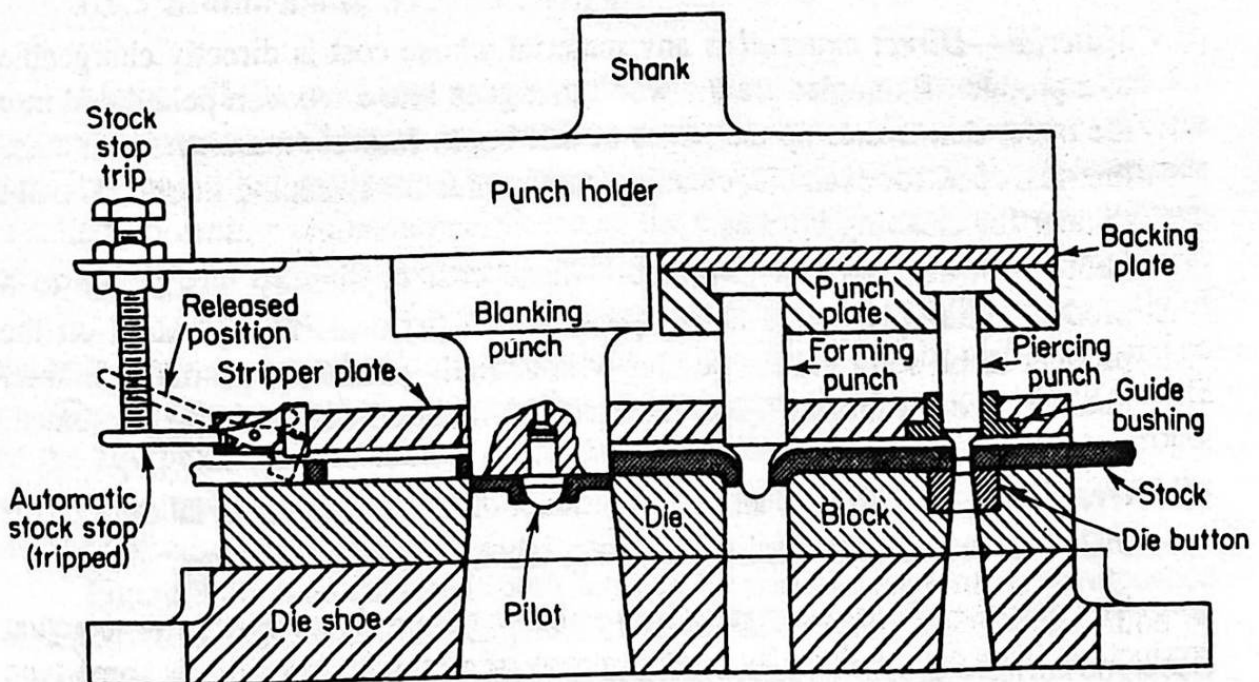


Fig. 3.23. Progressive die example. [Taken with permission, from Lawrence E. Doyle et al., *Manufacturing Processes and Materials for Engineers*, 2nd ed. (Englewood Cliffs, N.J.: Prentice Hall, 1969).]

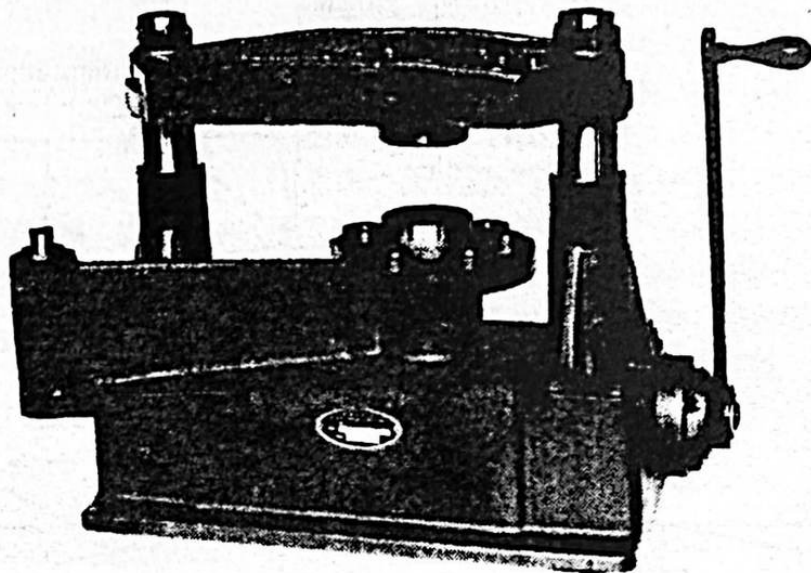


Fig. 3.24. Jig to line-bore six holes in a master rod for an aircraft engine.

example of a jig. Jigs and fixtures must be designed for each unique application. Consequently, the designer must be highly creative and knowledgeable in the area.

3.5.2 Cost Estimating

Of primary importance to the success of any manufacturing endeavor is cost estimating. *Cost estimating* is the determination of the total manufacturing cost and consists of the following three items:

- *Material*—*Direct material* is any material whose cost is directly chargeable to a product. Examples are the wood that goes into a wooden pencil and into the paper that makes up the pages of this book. *Indirect material* is all other material cost, for example, cleaning compounds for sweeping floors and paint thinner for cleaning brushes.
- *Labor*—*Direct labor* is any labor whose cost is directly chargeable to a product. Examples are a lathe operator who performs an operation on the product to be sold and the painter who actually paints the product. *Indirect labor* is all other factory labor, for example, maintenance personnel, foremen, and inspectors.
- *Overhead*—Overhead is all cost of production other than direct labor or direct material. Indirect material and indirect labor are overhead items.

The purpose of cost estimating is to estimate the product's cost prior to actual production. This can be done by several means or methods, but usually some type of planning sheet is used. A *planning sheet* is a form designed for detailed processing of the product through the various stages of manufacture, including all equipment and tool requirements. Costs are added to the planning sheet.

The actual form used varies widely between companies. Usually, enough detail is provided to be able to generate the routing sheets. Figure 3.25 is one

**Automotive Vehicles
3,000 Miles or 3-Months
Preventive Maintenance Service**

Vehicle No. _____

Date	Make and Model			Miles		
Service:	Fill Battery	Radiator Liquid	Fill Windshield Washer.	Check Engine Oil Level	Tire Pressure	Anti Freeze
Item No.	Description	Status	Corrective Action Including Material	Initials	Hours	
A. Interior						
1.	Foot and Hand Brake					
2.	Ignition Switch					
3.	Starter					
4.	Clutch Pedal					
5.	All Dash Instruments					
6.	Horn					
7.	Windshield Wipers and Washers					
8.	Gear Shift Loose or Broken					
9.	Steering Loose Motion					
10.	Fire Extinguisher Check Seal and Bracket					
B. Exterior						
1.	Mirror and Brackets					
2.	Doors and Latches					
3.	Chock Blocks					
4.	Tires—RR LR RF LF					
Underneath						
Check Exhaust System						
Under Hood						
Check Leaks, Hoses, Water, Fuel, Oil and Brake Fluid						
Lights						
Check all Lights and Reflectors						

Foreman Signed _____ Date _____

Fig. 3.26. Automotive vehicles 3,000 miles or 3-month preventive maintenance service.

3.5.4 Packaging Systems

Packaging is the last phase of what we have been calling production—manufacturing engineering. Here, the packaging of the finished product must be designed to protect it during transit and to ensure that the product will arrive undamaged. The package designer must be familiar with the various materials for packaging,

the techniques of package design, and the cost of various types of packages. He or she must also know how to set up workstations in order to obtain maximum effectiveness and efficiency.

Packaging is becoming a large area in itself. This may be seen by observing the interest in packaging programs throughout the country and the high demand for good package engineers. "The best package is no package" is an often heard quote. This implies that when economically feasible the product should be designed to withstand the perils of handling during shipment. When this becomes infeasible, a protective package is required. Other purposes of packaging include ease and aesthetics of product display.

3.6. Example

The Wheatley Company, a progressive pump and valve manufacturing company located in Tulsa, Oklahoma, has decided to start production on a new check valve, pictured in Figure 3.27.

The steps in the process engineering of this product are depicted below.⁴ Only the bill of material, operation process chart, and a sample routing sheet are shown. The bill of material (highly simplified) is shown in Figure 3.28. The operations process chart is developed next. It is shown in Figure 3.29 using fictitious machine numbers.

After the operations process chart has been prepared, the planning sheet is usually completed. Here, all details on material handling equipment, cost and time estimates, and so on are given. Next, a routing sheet is prepared for each part requiring processing. The routing sheet for the body only is shown in Figure 3.30. The form is such that it can be reproduced later and used as a production order, which instructs the shop to initiate production. Separate routing sheets would also be prepared for the cover, clapper, and arm, but they are not shown here.

The next step is to be sure that all necessary tools, jigs, and fixtures are available, design them when necessary, and issue production orders. Of course, packaging and maintenance systems must also be designed.

3.7. Computer Applications

Computers have changed the nature of production processes in many ways. They are often used to control the production process. For example, a minicomputer can be programmed to automatically adjust the process through a feedback mechanism. This is common in some process industries, such as paper mills. Although numerical control (the use of punched tape to control the operation of a machine) is not a

⁴Routing information was supplied by the Wheatley Company. The forms used are those of the authors, not the company. The authors are deeply indebted to the Wheatley Company for their aid and generosity in providing this example.

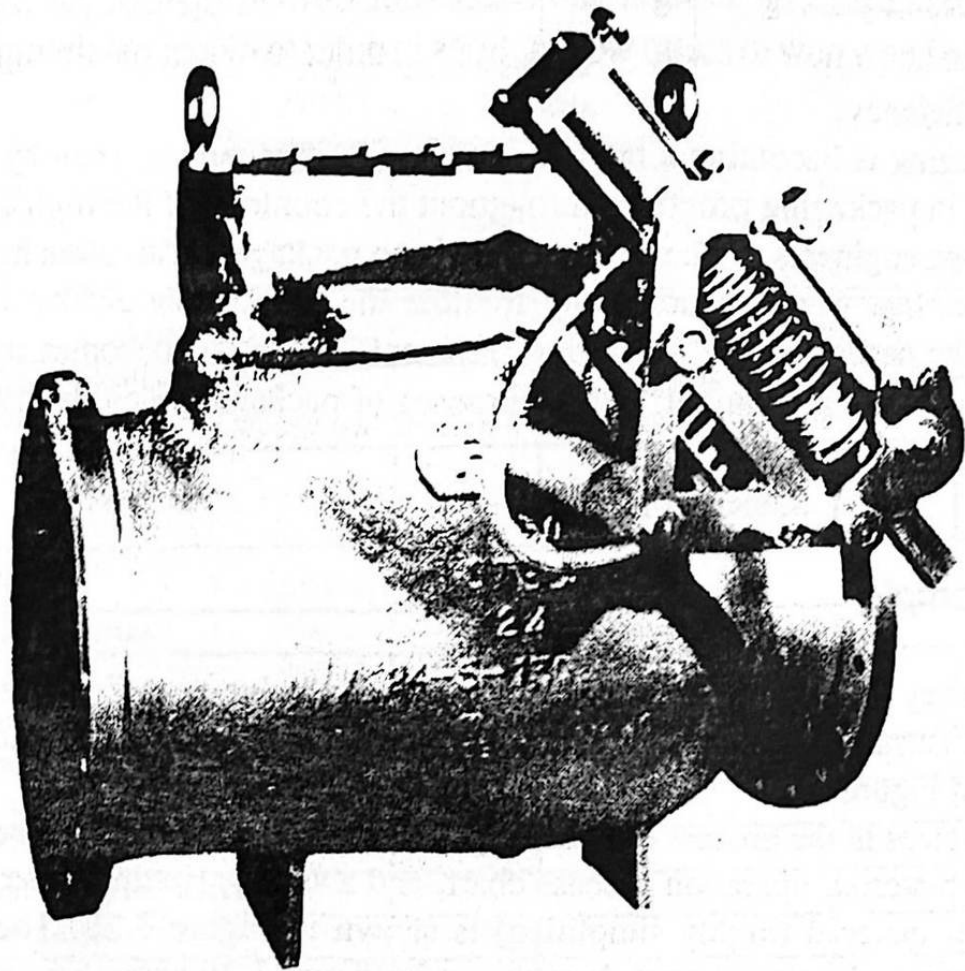


Fig. 3.27. Check valve.

Bill of Material

Model: Check Valve Type X Design Date _____ Sheet 1 of 1

Item	Quantity	Description	Source	Part No.
1	1	Body	Manufacturing	227
2	1	Cover	Manufacturing	275
3	8	Stud	Purchasing	271
4	8	Nut	Purchasing	243
5	1	Gasket, RTJ	Purchasing	353
6	1	Seat	Purchasing	236
7	1	O-Ring	Purchasing	201
8	1	O-Ring (Seat Backing)	Purchasing	301
9	1	Clapper	Manufacturing	237
10	1	Washer	Purchasing	220
11	1	Pin, "Roll Pin"	Purchasing	406
12	1	Arm	Manufacturing	221
13	1	Arm Pin	Purchasing	406
14	1	Plug, Arm Pin	Purchasing	420
15	1	Name Plate	Purchasing	282
16	2	Screw, Drive	Purchasing	320

Fig. 3.28. Bill of material for check valve.

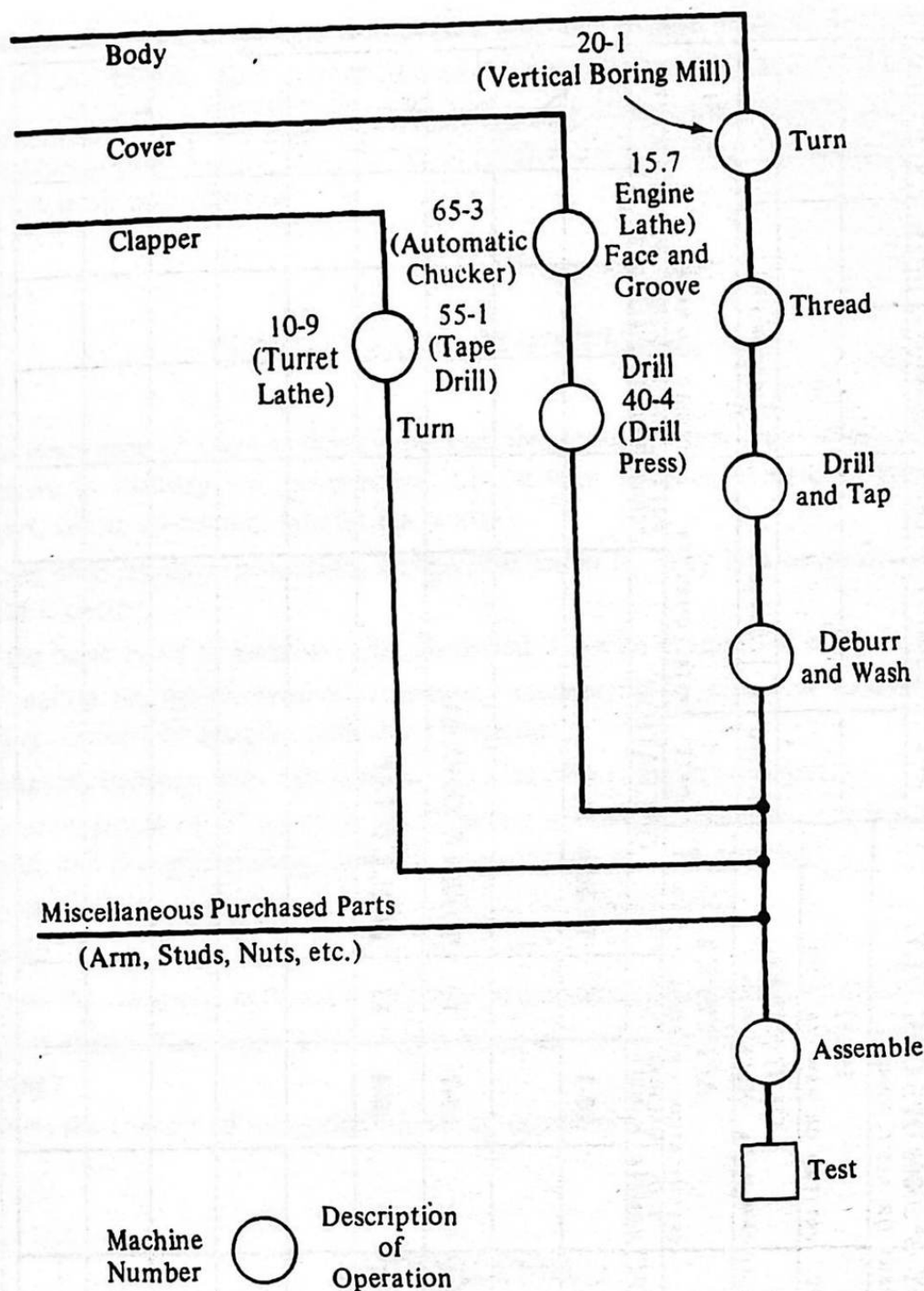


Fig. 3.29. Operations process chart for check valve.

computerized process, computer programs do exist that will translate the part's requirements into a numerical control language and automatically punch the tape (a very tedious process when done by hand).

It is often helpful to use a computer to set the parameters of the production process. At the extreme, the computer can be told what the part is, its dimensions, requirements, and so on, and it can completely design the process, including the order of operations, all feeds and speeds, and so forth. This last application is currently being researched and has, in fact, been done for some parts by using a mathematical programming formulation (mathematical programming will be discussed in Chapter 14).

→ Industry 4.0

The trend today is to use computers and production control techniques to integrate islands of automation in search of a totally automated factory. This means that computers will schedule, sequence, and control machine centers, robots, and automatic machining centers. The future is bright and unlimited for computer usage in manufacturing engineering.